



## Windows PC infected

More than 10,000 machines could be infected by NSA's advance backdoor.  
<http://dgit.in/DoublePulsar>



## McDonald's 'Mulan'

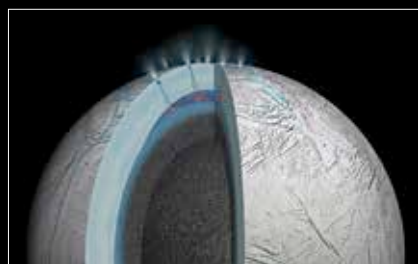
The internet has been flooded with McDonald's two decade old Szechuan sauce. Why?  
<http://dgit.in/McDviral>

Space Age

the surface of another planet. It was also discovered that the moon might also harbour a liquid water/ammonia ocean underneath the surface. With the presence of frozen hydrocarbons and water, it's quite likely in the future (millions of years later or even more), when the Sun's temperature increases, for life to originate there.

### The origin of life on Enceladus

Previously, Cassini had discovered icy water geysers on Enceladus, one of Saturn's moons. Cracks in the icy crust of the moon allowed water vapour, simple organic chemicals and ice particles to spew out. Spewing at about 400 metres per second, it forms a plume over the surface of the moon. Beneath the icy



Enceladus: Possible Hydrothermal Activity (Artist's Concept)

crust, the moon consists of a giant global ocean of water. This essentially means that there's a layer of water between the icy crust and rocky core. It gets interesting since this layer is made of salt water, something which is similar to what we have on Earth. In October 2015, Cassini did a close flyby through the plumes to analyse the accurate chemical composition of the gases. With the help of the on-board Ion and Neutral Mass Spectrometer, it was able to detect molecular hydrogen, leading to a possibility of hydrothermal reactions occurring between the ocean and rocky core. A similar water-rock interaction is also found on Earth in the form of hydrothermal vents on the bottom of the ocean floor. The release of molecular hydrogen could be an energy source for microbes who consume hydrogen to give out methane. Now that we know about the presence of hydrogen, we need to determine whether the

same hydrothermal process is happening in the ocean. This makes Enceladus another prime candidate to support life.

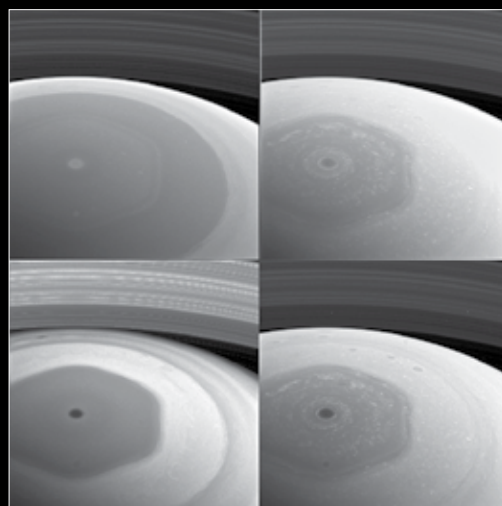
### The beginning of the end

You might wonder why crash the spacecraft rather than letting it float in the emptiness of the outer solar system? The primary reason for this step is to avoid collision into Titan or Enceladus, both Saturn's moons which apparently have signs of harbouring a fundamental form of life. The second factor to influence this move was the opportunity to study Saturn more closely, especially the uncharted and possibly empty region between the rings and the planet. In this way, we'll also get to study Saturn's atmosphere like never before. Essentially, every spiral around Saturn will generate some amount of scientific data that can be relayed back to Earth. The orbiter has been doing rounds of the ringed gas giant since its arrival to conduct research with a safer approach of avoiding ring debris. Because the probe has nearly emptied its fuel, it's worth the risk to get even closer to the planet.

Travelling at a staggering speed of 122,000 kph (76,000 mph), the spacecraft will complete 22 orbits around Saturn, covering the inner gap of 1,930 kilometres (1,200 miles). Since it's currently on a different trajectory, Cassini will use Titan's gravity to slingshot itself into Saturn's orbit through a gap in the rings. Until the Cassini mission, there has been a lot of mystery surrounding Saturn's rings. Believed to be a static system of boulders circling the planet, it was discovered that it's quite a volatile and active system of collisions. Further research will enable us to study more



Saturn, Approaching Northern Summer



Saturnian Hexagon Collage

about the science behind collisions between bodies in outer space. We'll also get more detailed information on how two or more bodies collide in space to form moons or even planets.

Saturn's magnetic and gravitational fields can be closely studied, giving us an idea of what lies beneath the gaseous atmosphere. Due to its close proximity, we'll receive detailed data for the first time on the composition of the gaseous clouds in its atmosphere, using the onboard mass spectrometer. Another mystery that has riddled astronomers and scientists for several years is the hexagonal cloud on the planet's north pole, something we might be able to demystify after this mission.

The final dive also poses a fatal risk. While travelling at such high speeds, impacts from even small particles might blow up the entire spacecraft. Cassini's ring-plane crossings will be oriented to shield off the vulnerable instruments behind the high-gain antenna. The team behind this mission are certain that the probability of experiencing a deadly impact is "less than 1 percent".

The final dive is going to be an emotional moment for the team that has worked on the Cassini-Huygens mission throughout its epic journey across the solar system. The mission was launched more than 19 years ago and the monumental journey is now getting closer to an end. Will we find life in any form other than Earth? The search continues...

Credit: NASA/JPL/Space Science Institute

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